

1. Поиск MAC-адреса для своего активного сетевого интерфейса (CLI || GUI) с помощью команды ipconfig/all. MAC-адрес находится в строчке Physical Address.

```
C:\Program Files (x86)\Microsoft Visual Studio\2022\BuildTools>ipconfig/all

Windows IP Configuration

Host Name . . . . . : LAPTOP-10LDD1GO
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
Description . . . . . : Killer E2600 Gigabit Ethernet Controller
Physical Address. . . . . : 08-8F-C3-02-79-AE
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-09
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter * 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 56-14-F3-FC-59-90
```

2. Запрос к узлу 192.168.88.1 с помощью команды ping.

```
C:\Program Files (x86)\Microsoft Visual Studio\2022\BuildTools>ping 192.168.88.1

Pinging 192.168.88.1 with 32 bytes of data:
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64
Reply from 192.168.88.1: bytes=32 time=2ms TTL=64
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64
Reply from 192.168.88.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.88.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

3. Физический адрес узла, к которому до этого был сделан запрос.

```
C:\Program Files (x86)\Microsoft Visual Studio\2022\BuildTools>arp -a
```

```
Interface: 192.168.56.1 --- 0x9
```

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

```
Interface: 192.168.88.25 --- 0xa
```

Internet Address	Physical Address	Type
192.168.88.1	74-4d-28-71-04-c8	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static